

Question 1

Given $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

(a) Let $u = \sqrt{a}x$. So $u^2 = ax^2$ and $du = \sqrt{a}dx$.

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-ax^2} dx &= \int_{-\infty}^{\infty} e^{-u^2} \cdot \frac{1}{\sqrt{a}} du \\ &= \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2} du \\ &= \sqrt{\frac{\pi}{a}} \end{aligned}$$

(b) Let $u = \frac{x-a}{b}$. So $u^2 = \frac{(x-a)^2}{b^2}$ and $du = \frac{1}{b}dx$.

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-\frac{(x-a)^2}{b^2}} dx &= \int_{-\infty}^{\infty} e^{-u^2} \cdot b du \\ &= b \int_{-\infty}^{\infty} e^{-u^2} du \\ &= b\sqrt{\pi} \end{aligned}$$

(c) First we simplify

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-ax^2+2bx} dx &= \int_{-\infty}^{\infty} e^{-a\left(x^2-\frac{2b}{a}x\right)} \\ &= \int_{-\infty}^{\infty} e^{-a\left(x^2-\frac{2b}{a}x+\frac{b^2}{a^2}-\frac{b^2}{a^2}\right)} \\ &= \int_{-\infty}^{\infty} e^{-a\left(x^2-\frac{2b}{a}x+\frac{b^2}{a^2}\right)+\frac{b^2}{a}} \\ &= \int_{-\infty}^{\infty} e^{-a\left(x-\frac{b}{a}\right)^2+\frac{b^2}{a}} \\ &= e^{\frac{b^2}{a}} \int_{-\infty}^{\infty} e^{-a\left(x-\frac{b}{a}\right)^2} \end{aligned}$$

Let $u = \sqrt{a}(x - \frac{b}{a})$. So $u^2 = a(x - \frac{b}{a})^2$ and $du = \sqrt{a}dx$.

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-ax^2+2bx} dx &= e^{\frac{b^2}{a}} \int_{-\infty}^{\infty} e^{-u^2} \cdot \frac{1}{\sqrt{a}} du \\ &= e^{\frac{b^2}{a}} \sqrt{\frac{\pi}{a}}\end{aligned}$$

(d) We use integration by parts. Let $u = x$ and $dv = xe^{-ax^2} dx$.

$$du = dx \quad v = \int xe^{-ax^2} dx$$

Let $w = -ax^2$ so $dw = -2ax dx$ and $dx = \frac{-1}{2ax} dw$. So we have

$$\begin{aligned}v &= \int \frac{-1}{2ax} \cdot xe^w dw \\ &= \frac{-1}{2a} e^{-ax^2}\end{aligned}$$

So in all,

$$\begin{aligned}\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= x \cdot \frac{-1}{2a} e^{-ax^2} \Big|_{-\infty}^{\infty} - \frac{-1}{2a} \int_{-\infty}^{\infty} e^{-ax^2} dx \\ &= 0 + \frac{1}{2a} \cdot \sqrt{\frac{\pi}{a}} \\ &= \frac{\sqrt{\pi}}{2\sqrt{a^3}}\end{aligned}$$

Question 2

(a) Given $u(x, 0) = \phi(x) = e^{-\frac{(x-a)^2}{b^2}}$.

$$\begin{aligned}
 u(x, t) &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} \phi(x - s\sqrt{4kt}) ds \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} e^{-\frac{(x-s\sqrt{4kt}-a)^2}{b^2}} ds \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{b^2+4kt}{b^2}s^2 + \frac{2(x-a)\sqrt{4kt}}{b^2}s + \frac{(x-a)^2}{b^2}\right)} ds \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{b^2+4kt}{b^2}\left(s^2 + \frac{2(x-a)\sqrt{4kt}}{b^2+4kt}s\right) - \frac{(x-a)^2}{b^2}\right)} ds \\
 &= \frac{e^{-\frac{(x-a)^2}{b^2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\frac{b^2+4kt}{b^2}\left(s^2 + \frac{2(x-a)\sqrt{4kt}}{b^2+4kt}s + \frac{(x-a)^2 4kt}{(b^2+4kt)^2} - \frac{(x-a)^2 4kt}{(b^2+4kt)^2}\right)} ds \\
 &= \frac{e^{-\frac{(x-a)^2}{b^2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\frac{b^2+4kt}{b^2}\left(s^2 + \frac{2(x-a)\sqrt{4kt}}{b^2+4kt}s + \frac{(x-a)^2 4kt}{(b^2+4kt)^2}\right) + \frac{(x-a)^2 4kt}{b^2(b^2+4kt)}} ds \\
 &= \frac{e^{-\frac{(x-a)^2}{b^2}} e^{\frac{(x-a)^2 4kt}{b^2(b^2+4kt)}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\frac{b^2+4kt}{b^2}\left(s + \frac{(x-a)\sqrt{4kt}}{b^2+4kt}\right)^2} ds \\
 &= \frac{e^{-\frac{(x-a)^2}{b^2} + \frac{(x-a)^2 4kt}{b^2(b^2+4kt)}}}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{\sqrt{-\frac{b^2+4kt}{b^2}}} \\
 &= \frac{e^{-\frac{(x-a)^2}{b^2} + \frac{(x-a)^2 4kt}{b^2(b^2+4kt)}}}{\sqrt{-\frac{b^2+4kt}{b^2}}}
 \end{aligned}$$

(b)

Question 3

(a)

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4kt}} \phi(y) dy$$
$$u(-x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(-x-y)^2}{4kt}} \phi(y) dy$$

(b)

$$-u(-x, t) = \frac{-1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(-x-y)^2}{4kt}} \phi(y) dy$$
$$-u(-x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(-x-y)^2}{4kt}} \phi(-y) dy$$

Question 4**Question 5**